



CENTRE FOR ADVANCED STUDIES

LUCKNOW, UTTAR PRADESH, INDIA

INSPIRING STUDENT INNOVATION

Deep Learning based identification of Bacteriosis and Quantification of Bacterial Spot Disease in Peach Crops.

A team of Centre for Advanced Studies developed an Artificial Intelligence based method for identification of disease using deep learning and quantification of bacteriosis in peach leaf using imaging method. Bacterial spot on peach leaf results in loss of fruit and the overall malaise of trees caused by recurrent defoliation. Leaves are the primary source of identification of plant diseases. The method performs bacteria detection from the peach leaf using different deep learning algorithms and an imaging method. In the imaging method, disease affected area is quantified and an adaptive operation is applied to a selected suitable channel of a color image. The method can be integrated with UAV device and connect with the user in the same local area network (LAN) to use as an early warning tool or a diagnostic model to help the farmer to keep eye on the field without being present there in person.

